Sampling Distributions and Bootstrap Distributions

Stat 141

In this worksheet, we'll consider the difference between **sampling distributions** and **bootstrap distributions**. As a brief reminder:

- In all cases, we are interested in estimating an unknown population parameter, p using a sample statistic, $\hat{p}$. In the real world, we calculate $\hat{p}$ using data from a sample of size n. We would like to understand how close our estimate, $\hat{p}$, might be to the unknown truth, p.
- A **sampling distribution** represents the distribution of all possible statistics (which could have come from every possible sample of size n from our *population*). We can't actually calculate a sampling distribution in the real world, because we only have one sample!
- A **bootstrap distribution** is a practical attempt to understand sampling variability given our single sample. Instead of showing the distribution of statistics from every possible sample from the population, it shows the distribution of statistics from every possible sample of size n from our *sample*.

Data: A Deck of Cards

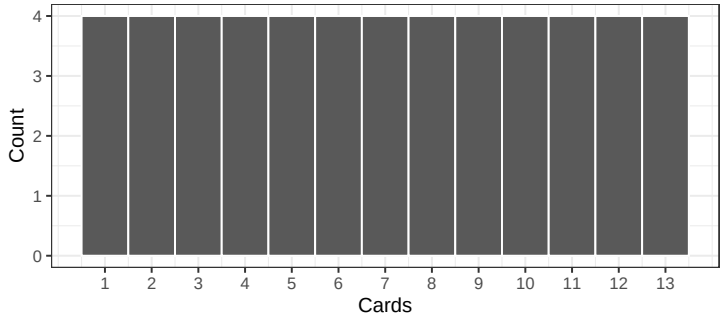
Yet again, we'll use the example of sampling from a deck of cards. As before, let's count:

- Aces as 1
- Jacks as 11
- Queens as 12
- Kings as 13

Below is code to create the "deck" virtually.

```
deck <- data.frame(cards = rep(1:13, each=4))
```

Fig. 1: Population Distribution



Q1: In Figure 1, why is the height of each bar 4? Describe this distribution.

- There are four cards of each value in a deck.
- The distribution has a "flat" shape, centered at 7 (mean = 7 and median = 7).
- The distribution has a "large" spread. Standard Deviation: 3.78

Sampling Distributions

When creating sampling distributions and bootstrap distributions, we will need a way to sample (from a population or a sample). With cards, it's possible to physically shuffle the deck and draw a "random sample". It's also possible to draw samples with code! Moving forward, we'll often use the `rep_sample_n()` function in R to sample. Here's an example:

```
set.seed(1) # ensures we all get the same "random" sample
single_sample <- deck %>% rep_sample_n(size = 10, replace = FALSE, reps = 1)
single_sample$cards
```

```
 [1]  1 10  1  9  6 11  4  5  9  6
```

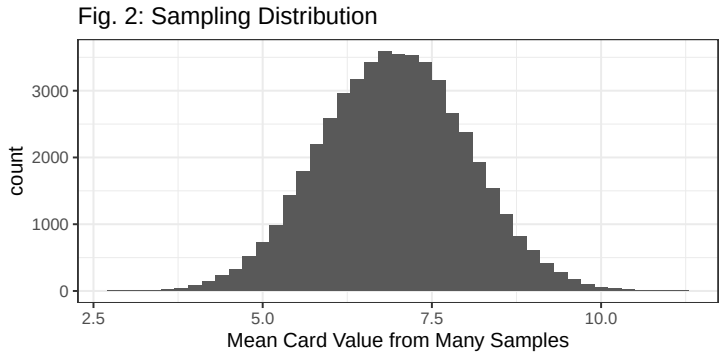
The code above took our `deck`, and took a simple random sample of `size = 10`. The additional arguments `replace = FALSE` means that we're sampling *without replacement*, and `reps = 1` means we took a single sample from the deck! We then print the `cards` in our `single_sample`.

Q2: Based on the code, which cards are in our sample? Use the cards to calculate our sample statistic (the sample mean) based on this sample.

- We have two Ace's (1's), a 4, 5, two 6's, two 9's, a 10, and a Jack (11).
- The sample mean is 6.2
  - Can calculate manually in this case
  - or could use `mean(single_sample$cards)`

The following code creates a ("snapshot" of the) sampling distribution:

```
deck %>%
  rep_sample_n(size = 10,
               replace = FALSE, reps = 50000) %>%
  group_by(replicate) %>%
  summarize(x_bar = mean(cards)) %>%
  ggplot(aes(x = x_bar)) +
  geom_histogram(binwidth = 0.2) + theme_bw() +
  labs(x = "Mean Card Value from Many Samples",
       title = "Fig. 2: Sampling Distribution")
```

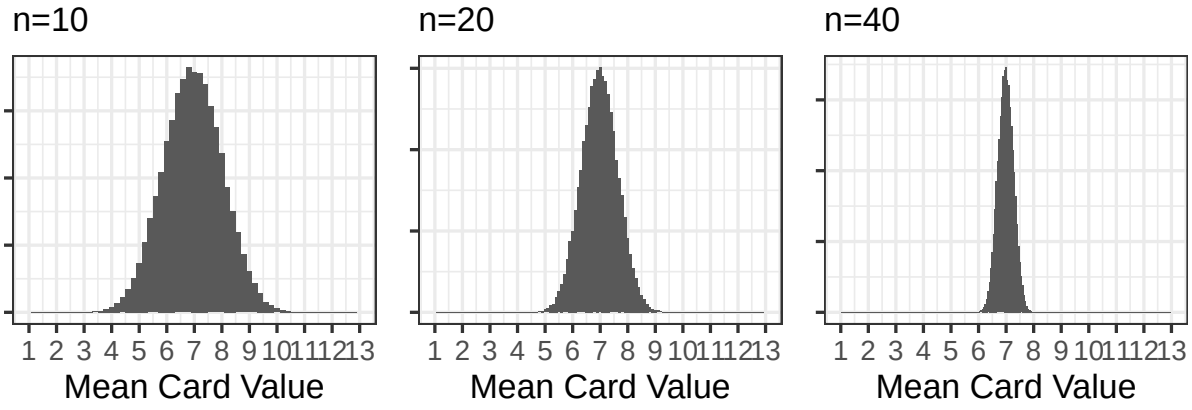


Q3: Based on the code above, how many cards are in each sample? How many different samples did we take to create the sampling distribution in Figure 2?

- There are 10 cards in each sample, since `size = 10`.
- We took 50,000 different samples, since `reps = 50000`.

Q4: Figure 3 (below) displays sampling distributions for samples of size n=10, n=20, and n=40. How are they similar and how are they different? Why do larger samples have sampling distributions with less variability?

Fig. 3



- Bell-shaped and centered at the true mean, 7.
- They differ in their spread: $n = 10$ distribution has the largest standard error; the $n = 40$ distribution has the smallest standard error.
- **Why?** If our sample size (n) is larger, we have more data and our sample should be "more representative" of the population.
- i.e., more data means a better glimpse at the true population, and better guesses about the population mean!

Bootstrap Distributions

The following code creates a single "bootstrap sample".

```
bootstrap_sample <- single_sample %>% rep_sample_n(size = 10, replace = TRUE,
                                                   reps = 1)
bootstrap_sample$cards
```

```
 [1] 11  6  9  1  5  6  1  1  9  9
```

Q5: In the code above, what are we sampling from (the population, or the single sample)? Are we sampling with replacement? What's our sample size?

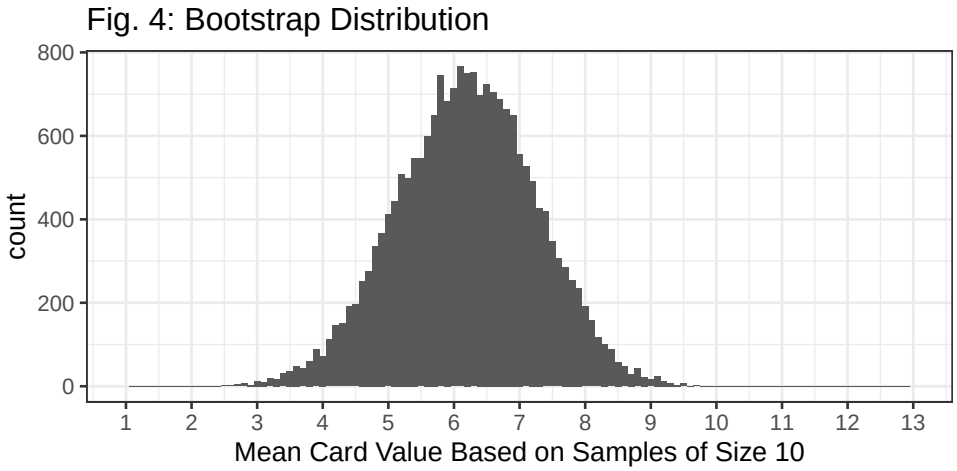
- We're sampling from the single sample (`single_sample`).
- We're sampling with replacement (`replace = TRUE`).
- Our sample size is still 10 (`size = 10`)

Q6: How would your answers to Q5 be different if we were talking about sampling for a sampling distribution?

- We sample from the **population**.
- We sample **without** replacement.
- Our sample size is **still 10**!

Next, we write code to calculate a bootstrap distribution:

```
single_sample %>% ungroup() %>% select(cards) %>%
  rep_sample_n(size = 10, replace = TRUE, reps = 20000) %>%
  group_by(replicate) %>%
  summarize(x_bar = mean(cards)) %>%
  ggplot(aes(x = x_bar)) + geom_histogram(binwidth = 0.1) +
  labs(x = "Mean Card Value Based on Samples of Size 10",
       title = "Fig. 4: Bootstrap Distribution") +
  theme_bw() +
  scale_x_continuous(breaks = 1:13, limits = c(1, 13))
```



Q7: Based on the code above, how many bootstrap samples are we taking to create the bootstrap distribution?

- 20,000 bootstrap samples because `reps=20000`.

Q8: Which sampling distribution from Figure 3 looks most like the bootstrap distribution in Figure 4? How specifically is it similar or different? Consider the shape, center, and spread of each distibution.

The $n = 10$ sampling distribution!

- **Shape:** All distributions look bell-shaped, so this doesn't distinguish them at all.

- **Center:** Bootstrap is centered at the sample mean (6.2); all sampling distributions are centered at the population mean (7). Again, this doesn't distinguish them.
- **Spread:** This does distinguish them! The spread of the bootstrap distribution looks similar to the spread of the sampling distribution with $n = 10$.

Q9: Thinking more generally, compare and contrast sampling distributions and bootstrap distributions.

- Sampling distributions and bootstrap distributions both help us conceptualize the distribution of a statistic, for the purpose of understanding plausible values for a parameter.
- Sampling distributions and bootstrap distributions should have similar spread (e.g., a similar standard deviation).
- Their means should also be similar, although a sampling distribution is centered at the true parameter, while a bootstrap distribution is centered at the sample mean.
- Sampling distributions are usually not possible to obtain (we can only take one sample from the population). Bootstrap distributions approximate a sampling distribution, and are super easy to obtain (especially with code).